

classification implementations

- Distributed fitness function capabilities for evolutionary programming
- Matrix and vector libraries
- Examples of all the above algorithms

Useful links

Mahout home: <http://mahout.apache.org/>

GitHub: <https://github.com/apache/mahout>

An introduction to Mahout by Grant Ingersoll: <https://www.ibm.com/developerworks/library/j-mahout/>

5) OpenNN: An open source class library written in C++ to implement neural networks

OpenNN (Open Neural Networks Library) was formerly known as Flood and is based on the Ph.D thesis of R. Lopez, called 'Neural Networks for Variational Problems in Engineering', at the Technical University of Catalonia, 2008.



OpenNN implements data mining methods as a bundle of functions. These can be embedded in other software tools using an application programming interface (API) for the interaction between the software tool and the predictive analytics tasks. The main advantage of OpenNN is its high performance. It is developed in C++ for better memory management and higher processing speed. It implements CPU parallelisation by means of OpenMP and GPU acceleration with CUDA.

The package comes with unit testing, many examples and extensive documentation. It provides an effective framework

for the research and development of neural networks algorithms and applications. Neural Designer is a professional predictive analytics tool that uses OpenNN, which means that the neural engine of Neural Designer has been built using OpenNN.

OpenNN has been designed to learn from both data sets and mathematical models.

Data sets

- Function regression
- Pattern recognition
- Time series prediction

Mathematical models

- Optimal control
- Optimal shape design

Data sets and mathematical models

- Inverse problems

Useful links

OpenNN home: <http://www.opennn.net/>

OpenNN Artnetics GitHub: <https://github.com/Artnetics/OpenNN>

Neural Designer: <https://neuraldesigner.com/>

6) Torch: An open source machine learning library, a scientific computing framework, and a script language based on the Lua programming language

Torch provides a wide range of algorithms for deep machine learning. It uses the scripting language LuaJIT, and an underlying C/CUDA implementation.



The core package of Torch is *torch*. It provides a flexible N-dimensional array or tensor, which supports basic routines for indexing, slicing, transposing, type-casting, resizing, sharing storage and cloning. The *nn* package is used for building neural networks.

Features

- It is a powerful N-dimensional array

- Has lots of routines for indexing, slicing and transposing
- Has an amazing interface to C, via LuaJIT
- Linear algebra routines
- Neural network and energy-based models
- Numeric optimisation routines
- Fast and efficient GPU support
- Embeddable, with ports to iOS and Android back-ends

Torch is used by the Facebook AI Research Group, IBM, Yandex and the Idiap Research Institute. It has been extended for use on Android and iOS. It has been used to build hardware implementations for data flows like those found in neural networks. Facebook has released a set of extension modules as open source software.

PyTorch is an open source machine learning library for Python, used for applications such as natural language processing. It is primarily developed by Facebook's artificial intelligence research group, and Uber's Pyro software for probabilistic programming has been built upon it.

Useful links

Torch Home: <http://torch.ch/>

GitHub: <https://github.com/torch>

7) Neuroph: An object-oriented neural network framework written in Java

Neuroph can be used to create and train neural networks in Java programs. It provides a Java class library as well as a GUI tool called easyNeurons for creating and training neural networks. Neuroph is a lightweight Java neural network, as well as a framework to develop common neural network architectures. It contains a well-designed, open source Java library with a small number of basic classes that correspond to basic NN concepts. It also has a nice GUI neural network editor to quickly create Java neural network components. It has been released as open source under the Apache 2.0 licence.

Neuroph's core classes correspond